SHIKHAR PANDAV

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PROFESSIONAL EXPERIENCE

Infosys Pune, India

Specialist Programmer (Power Programmer) *July 2024 – April 2026*

E-commerce Client — Order Processing (Python, FastAPI, RabbitMQ, Prometheus)

- **Re-architected** the order-processing path from sync handoffs to an at-least-once **RabbitMQ** pipeline with **idempotency-key dedup** and a **dead-letter queue**, eliminating duplicate orders and cutting **p95 latency 30%**.
- **Diagnosed** peak-traffic queueing as async **connection-pool saturation** via **Prometheus** histograms; rightsized pool limits and timeouts, reducing **p99 queue wait 40%** without adding replicas.
- **Instrumented Grafana** dashboards and alerts on queue depth, consumer lag and DLQ rate, cutting **triage time 25%**.
- **Authored pytest** integration suites for retry and DLQ paths, lifting pipeline coverage to **80%+** as a **CI/CD** merge gate.

Banking Client — Transactional APIs (Redis, gRPC/Protobuf, Microservices)

- **Designed** a **Redis** read-through cache for a high-traffic transactional API with tenant-scoped keys and per-endpoint TTLs, cutting **database read load 45%** and **p95 latency 35%**.
- **Migrated** inter-service REST/JSON to **gRPC/Protobuf**, shrinking payloads **40%** and catching contract breaks at build time via generated stubs and CI schema checks.
- **Standardized** structured logging with **correlation IDs** across **6 microservices**, enabling end-to-end tracing and reducing debugging time **30%**.

TECHNICAL PROJECTS

FORGE – Self-Hosted LLM Gateway & Agentic CLI

Backend Developer | Python, FastAPI, Redis, PostgreSQL, MCP *June 2026 – Present*

- **Developed** an **OpenAI-compatible LLM gateway** across **3 providers** with per-model **circuit breaking** and backoff failover, recovering from provider outages in **under 2s**.
- **Eliminated** race conditions in distributed **token-bucket rate limiting** with an atomic **Redis Lua** script, holding **0 over-limit admits** under **500 concurrent requests**.
- **Metered** tokens, cost and latency for **100%** of requests in an append-only **PostgreSQL** ledger, surfaced via a live metrics dashboard.
- **Integrated MCP** tool servers over **2 transports** (stdio, HTTP/SSE) via a fail-isolated plugin host, gated by a **5-mode approval engine**.
- **Shipped** a golden-task **LLM eval harness** and **55 automated tests** in **GitHub Actions CI**. GitHub

MyTube – Scalable Full-Stack Video Infrastructure

Full Stack Developer | Next.js, PostgreSQL, Drizzle, Mux *May 2025 – July 2025*

- **Engineered** a **Mux** adaptive-bitrate streaming pipeline supporting **10,000+** concurrent viewers with sub-2s buffering.
- **Accelerated** nested queries **50%** with **PostgreSQL** recursive **CTEs** via **Drizzle ORM**. GitHub | Website

TECHNICAL SKILLS

Languages: Python, C++, TypeScript, JavaScript, SQL, C#
Backend: FastAPI, Django, asyncio, SQLAlchemy, gRPC/Protobuf, RabbitMQ, Redis, PostgreSQL, REST APIs
DevOps & Observability: Docker, GCP, GitHub Actions CI/CD, Prometheus, Grafana, Pytest, Git, Linux
AI & GenAI: LLM Gateways, MCP, Agent Evals, RAG, LangChain, LangGraph, Vector Databases (ChromaDB)

ACHIEVEMENTS

- **LeetCode Guardian** — peak **2250** (top **1%**), **1,500+** problems, **Global Rank 207** / **21,482** in a Weekly Contest.
- **Codeforces Expert** — peak rating **1633**.

EDUCATION

KIET Group of Institutions, AKTU Delhi-NCR, India

Bachelor of Technology in Computer Science and Engineering *August 2020 – June 2024*